

PUBPOL 5390 Homework 2

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Draft date: 03 November 2025

1 Question 1

1.0.1 Cross-Validation

Within the tidymodels framework (the R analogue to scikit-learn), there are no defaults for cross-validation. Instead, I wrote my own function a la Mullinathan and Spiess, to run the models in an automated way. These functions allow the grid size for parameters to be searched over and the number of folds for cross-validation. For all models, I set the number of folds to 10, the grid size to 30 or 50 depending on whether the penalty is L1 or L2. The goodness of fit measure that I use is RMSE.

1.0.2 Model Performance

I use three different covariate sets that layer controls in each model and change the functional form. The first model includes linear forms of relevant predictors (e.g., mother's age and gestation). The second and third incorporate non-linear terms as well as expanded covariates. The majority of the improvement in predictive performance in richer models is obtained from gestation weeks.

Overall, the results, in Table 1, indicate that the best model in terms of predictive performance is a LASSO regression that selects 30 predictors out of a candidate set of 39 and achieves an out-of-sample RMSE of 437. Across ridge and elastic net, the RMSE is slightly worse but not by that much. In general, each of the second feature set models achieves similar results. The third feature set, which is the richest model, performs the

Table 1: Comparison of Predictive Performance Across Models and Feature Sets

Model	in_sample_sd_y	oo_sample_sd_y	is_rmse	oos_rmse	n_predictors	alpha	lambda
OLS	605.83	569.06	483.14	488.60	4.00		
OLS	605.83	569.06	452.13	475.79	39.00		
OLS	605.83	569.06	141.80	910.57	541.00		
LASSO	605.83	569.06	483.25	487.49	4.00	1.00	6.00
LASSO	605.83	569.06	454.60	473.38	30.00	1.00	9.00
LASSO	605.83	569.06	418.56	503.89	94.00	1.00	30.00
Ridge	605.83	569.06	483.58	487.02	5.00	0.00	0.00
Ridge	605.83	569.06	453.11	473.73	40.00	0.00	50.00
Ridge	605.83	569.06	164.23	670.86	542.00	0.00	50.00
Elastic	605.83	569.06	483.24	487.53	4.00	0.75	7.00
Elastic	605.83	569.06	455.63	473.41	29.00	0.75	14.00
Elastic	605.83	569.06	438.04	500.96	59.00	0.75	46.00

Table 2: Replication of Mullinathan Spiess Table 1

Model	Features	in_sample_rmse	oo_sample_rmse	is_r2	oos_r2	P1	P2	P3	P4	P5
OLS	Rep	0.84	0.86	0.42	0.40	0.00	0.00	0.00	0.00	0.00
OLS	KS	0.81	0.83	0.46	0.43	9.49	26.26	-287.43	14.51	13.45
LASSO	Rep	0.84	0.86	0.42	0.40	-0.02	0.28	-9.77	0.07	-0.04
LASSO	KS	0.81	0.83	0.46	0.43	9.45	26.51	-299.18	14.62	13.50
Elastic	Rep	0.84	0.86	0.42	0.40	-0.02	0.25	-9.32	0.04	-0.05
Elastic	KS	0.81	0.83	0.46	0.44	9.46	26.52	-301.08	14.66	13.48

worst across models. Though, the LASSO model shrinks the feature set from 542 to 94, which reduces the RMSE. Even with shrinkage, it still performs poorly.

2 Question 2

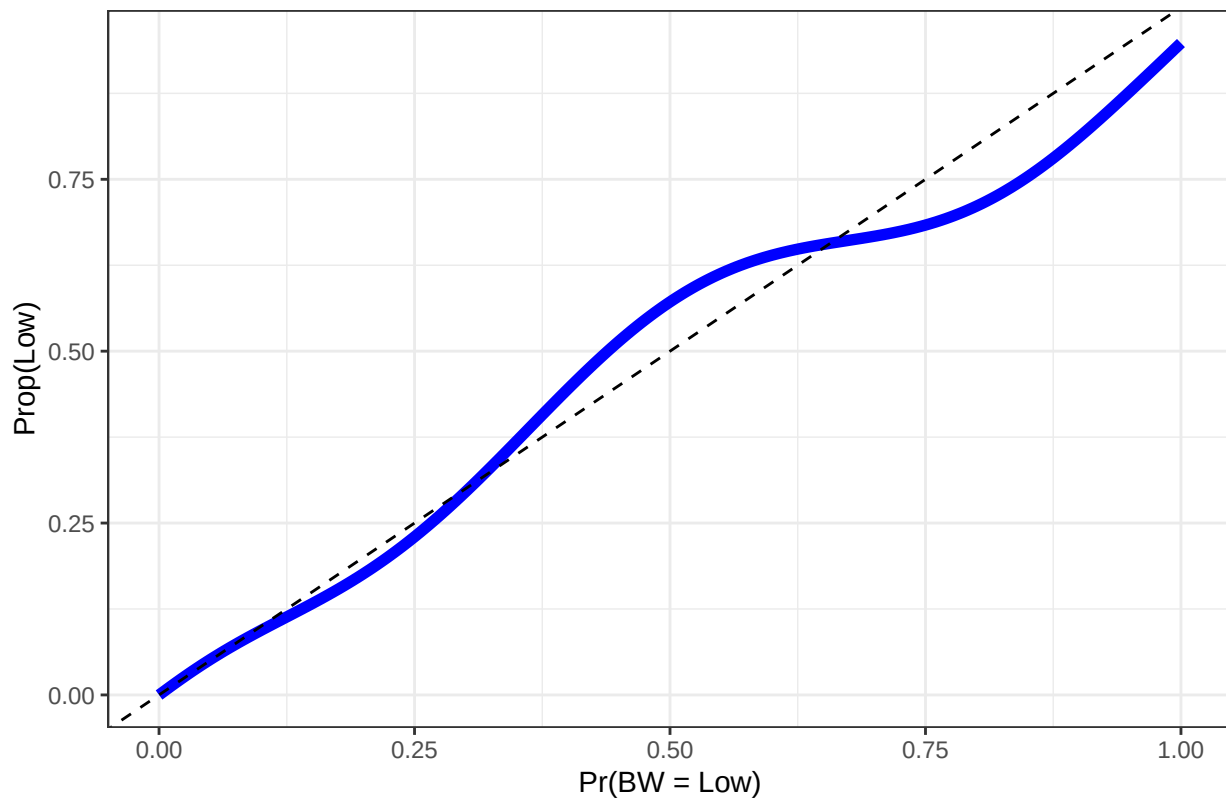
The second table describes the results of Replication and Kitchen Sink Regressions for OLS, LASSO, and Elastic Net. Across models, the replication and kitchen sink regressions do about the same. The best performing model is the Elastic net, but it is only by a small margin (e.g. 44 v 43 R2). By House Price Quintile (P1-P5) each of the kitchen sink models achieve large performance improvements in the 3rd quintile for all models.

3 Question 3

4 b.

4.1 Calibration Plot

Calibration Plot (Small)



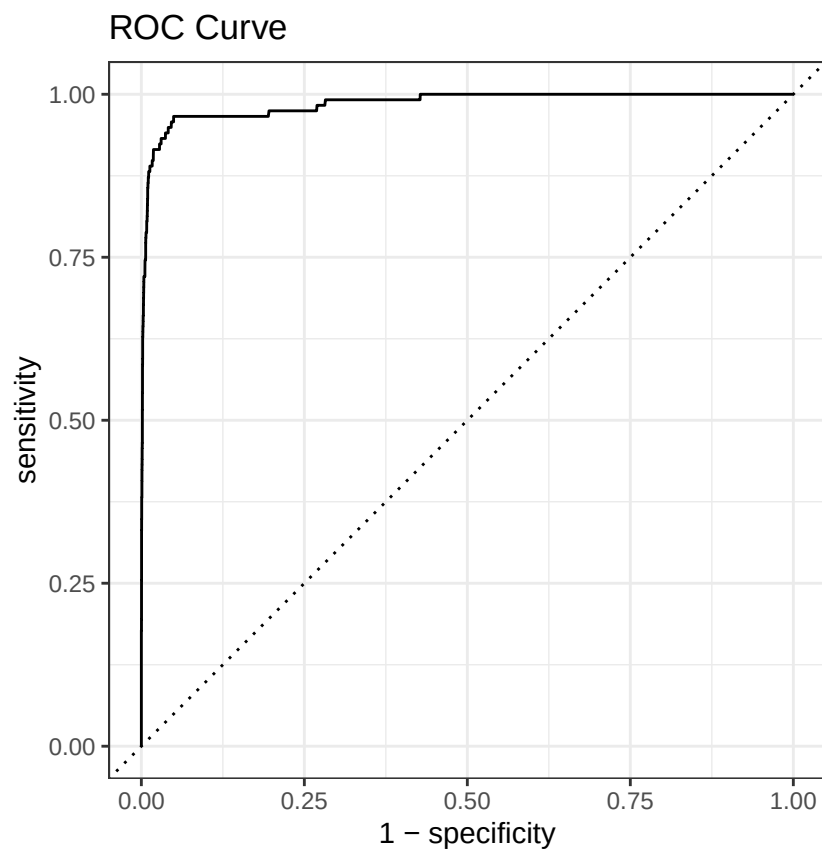
The calibration plot displays the predicted probabilities of low-birthweight on the X axis and the share of observations classified as low-birthweight on the Y axis for each bin of predicted probabilities based on a local polynomial regression using a bandwidth of .3 and a uniform kernel. What we can broadly see is that at higher values of predicted low-birthweight, the model overpredicts the number of low-birthweight babies, while the true rates of low birthweight are lower than the 45 degree line. In the middle of the distribution of predicted probabilities, around .25-.6, the model tends to underpredict birthweight.

4.2 Confusion Matrix

```
##           Truth
## Prediction  low normal
##    low      67    20
##    normal   51 11400
```

For the confusion matrix, 67 true positives, 11420 true negatives, 51 false negatives, and 20 false positives.

4.3 ROC Curve



There is not a direct ROC curve in-

terpretation.

Table 3: Probit GOF (Feature set 1)

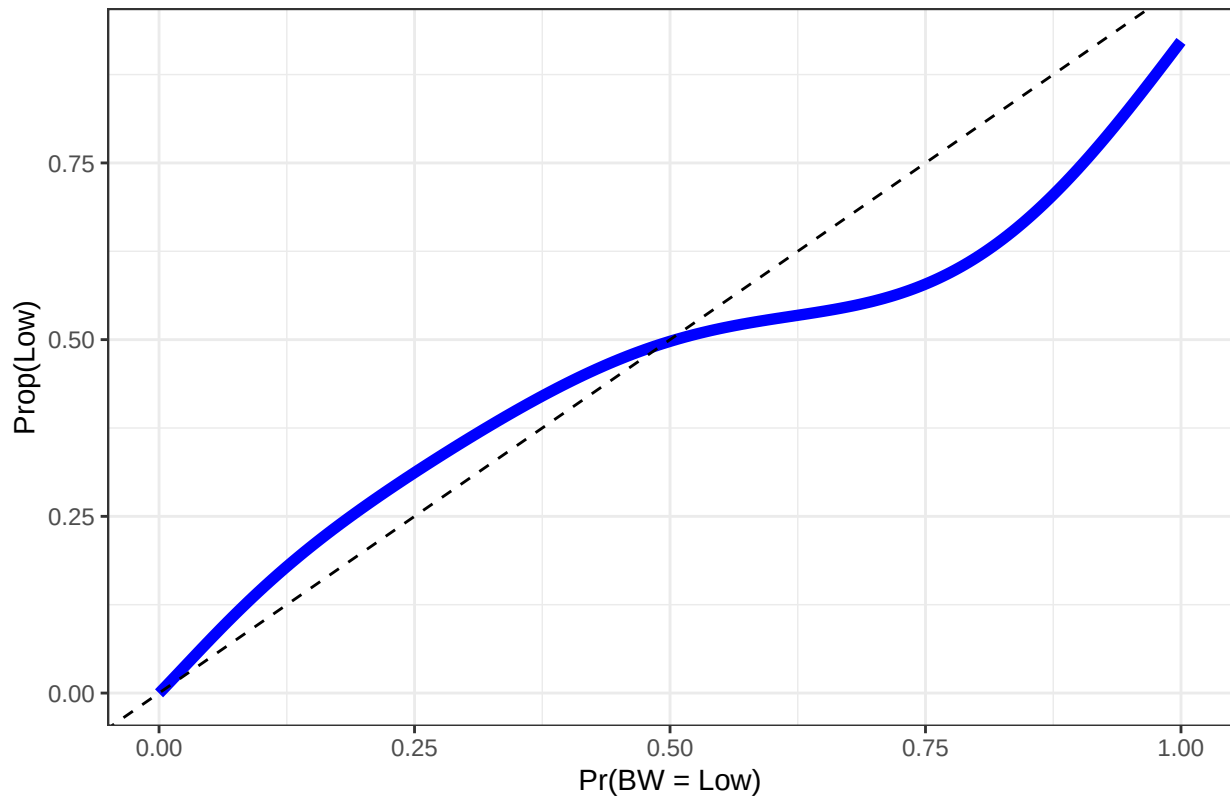
.metric	.estimator	.estimate
accuracy	binary	0.99
precision	binary	0.77
sens	binary	0.57
roc_auc	binary	0.99

4.4 AUC

The Probit GOF (Set 1) describes the results of various goodness of fit statistics. The first is the accuracy, which measures the share of correct classifications out of total classifications. The number correct is .99. Second is precision, which refers to the proportion of the models positive classifications that are actually positive, which has. value of .77. The sensitivity is the true positive rate, which is the probability of a positive result given the case being positive, which is .57. Finally, I report the AUC which is .99. Relative to a model classifying at random, the model does quite well.

4.5 c.

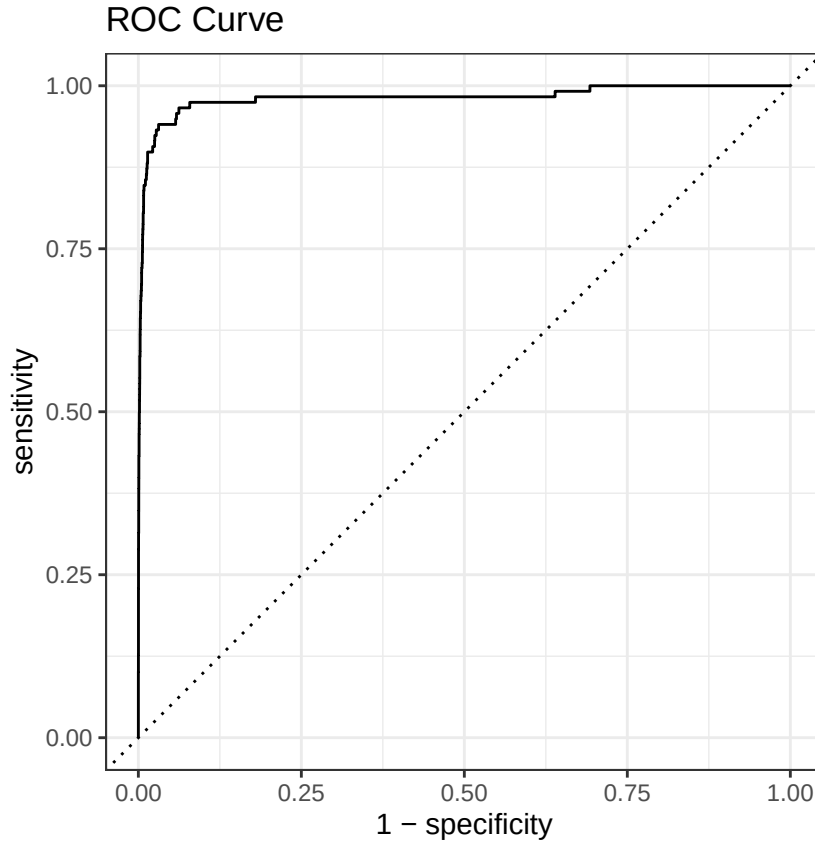
Calibration Plot (Large Feature Set)



```
##           Truth
## Prediction  low normal
##    low      64    23
##    normal   54 11397
```

Table 4: Probit GOF (Feature set 3)

.metric	.estimator	.estimate
accuracy	binary	0.99
precision	binary	0.74
sens	binary	0.54
roc_auc	binary	0.98



For the confusion matrix, 64 true positives, 11397 true negatives, 54 false negatives, and 23 false positives.

The Probit GOF (Set 3) table describes the results of various goodness of fit statistics. The first is the accuracy, which measures the share of correct classifications out of total classifications. The number correct is .99. Second is precision, which refers to the proportion of the models positive classifications that are actually positive, which has a value of .74. The sensitivity is the true positive rate, which is the probability of a positive result given the case being positive, which is .54. Finally, I report the AUC which is .98. Relative to the small feature set, this model does a little bit worse at the true positive rate (sensitivity) and the AUC. But it

is not dramatic.